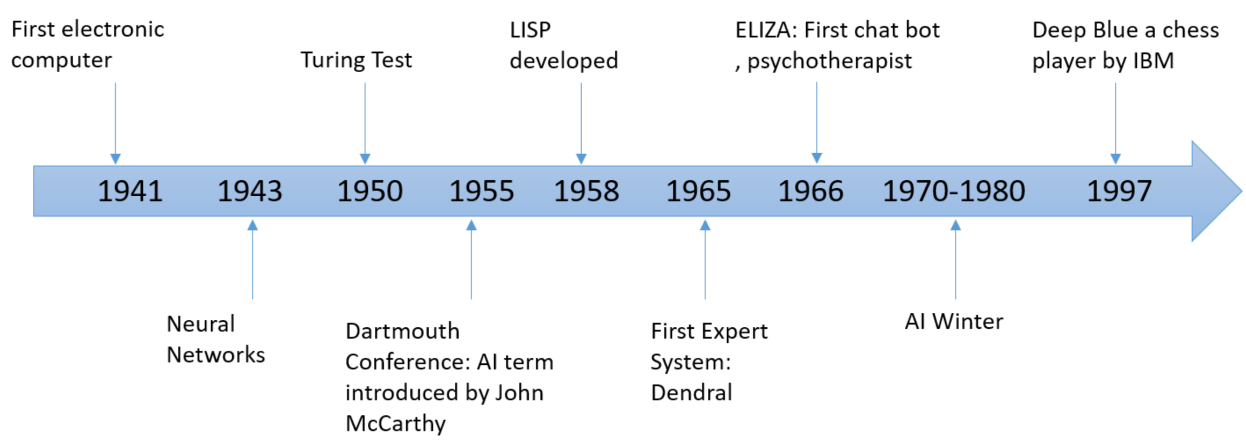
Artificial Intelligence (CSC-203)

Objectives:

- **AI History**
- **Expert Systems Components**
- **Expert System Limitations**
- **Expert System Shell**

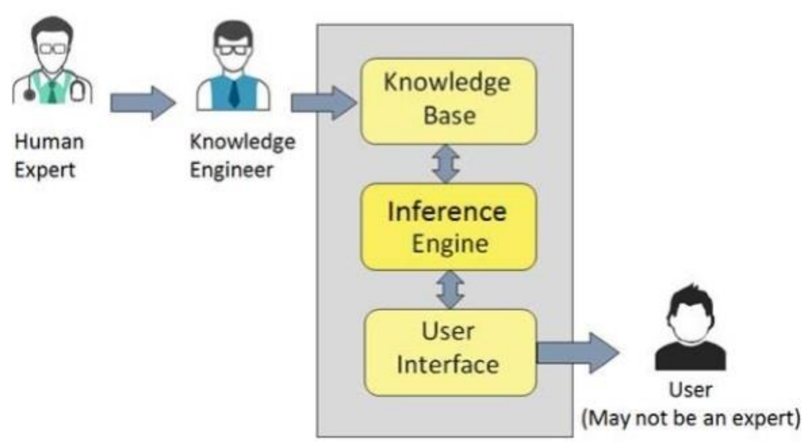
AI History



What is Expert System?

Expert System is an interactive and reliable computer-based decision-making system which uses both facts and heuristics to solve complex decision-making problems. It is considered at the highest level of human intelligence and expertise. The purpose of an expert system is to solve the most complex issues in a specific domain.

Components of an Expert System



1. Knowledge Base

It contains domain-specific and high-quality knowledge.

Knowledge is required to exhibit intelligence. The success of any Expert System majorly depends upon the collection of highly accurate and precise knowledge.

What is Knowledge?

The data is collection of facts. The information is organized as data and facts about the task domain. **Data, information,** and **past experience** combined together are termed as knowledge.

Components of Knowledge Base

The knowledge base of an ES is a store of both, factual and heuristic knowledge.

- **Factual Knowledge** − It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
- **Heuristic Knowledge** − It is about practice, accurate judgement, one's ability of evaluation, and guessing

Knowledge representation

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

2. Inference Engine

The Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.

Use of efficient procedures and rules by the Inference Engine helps in deducting a correct, flawless solution.

To recommend a solution, the Inference Engine uses the following strategies −

- Forward Chaining
- Backward Chaining

Forward Chaining:

It is a strategy of an expert system to answer the question, **"What can happen next?"**

Here, the Inference Engine follows the chain of conditions and derivations and finally deduces the outcome. It considers all the facts and rules, and sorts them before concluding to a solution.

This strategy is followed for working on conclusion, result, or effect. For example, prediction of share market status as an effect of changes in interest rates.

Backward Chaining:

With this strategy, an expert system finds out the answer to the question, **"Why this happened?"**

On the basis of what has already happened, the Inference Engine tries to find out which conditions could have happened in the past for this result. This strategy is followed for finding out cause or reason. For example, diagnosis of blood cancer in humans.

3. User Interface

User interface provides interaction between user of the ES and the ES itself.. The user of the ES need not be necessarily an expert in Artificial Intelligence.

The explanations of ES may appear in the following forms.

- Natural language displayed on screen.
- Verbal narrations in natural language.
- Listing of rule numbers displayed on the screen.

4. Knowledge Acquisition

The term knowledge acquisition means how to get required domain knowledge by the expert system. The entire process starts by extracting knowledge from a human expert, converting the acquired knowledge into rules and injecting the developed rules into the knowledge base.

The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base.

The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills. He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

Expert Systems Limitations

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include.

- Limitations of the technology
- Difficult knowledge acquisition
- High development costs
- ES are difficult to maintain (in terms of emerging knowledge and in terms of cost)
- Unable to make a creative response in an extraordinary situation

- Errors in the knowledge base can lead to wrong decision
- Each problem is different therefore the solution from a human expert can also be different (conflict in experts)

Expert System Shells

It is a software containing no knowledge (but have mechanism for representing the knowledge and Inferencing on it) used to develop expert systems. It is like an expert system containing all the modules except knowledge. Example is OPS-5 (Official Production system).

Expert system Examples

- DENDRAL
- MACSYMA
- MYCIN
- PROSPECTOR
- HEARSAY-II
- PUFF, GUIDON, NEOMYCIN
- HASP
- INTERNIST
- XCON
- DRILLING ADVISOR

Expert systems have not been around for very long. The following list places some of the best known systems in context:

- Beginnings (1965 – 1970) : DENDRAL, MACSYMA
- Prototypes (1970 - 1975) : MYCIN, PROSPECTOR, HEARSAY
- Experimenting (1975 – 1981): PUFF, GUIDON, INTERNIST
- Commercial Systems (1981- onwards): XCON, Drilling Advisor